

Phase 7 Runbook

Use this runbook for the modular operator-discovery track after activating the shared environment:

```
conda activate Python3_14
```

Preparation

Refresh the TSP benchmark manifest so the phase-6 descriptors and the phase-7 validation families are present:

```
python prepare_tsp_benchmarks.py
```

Smoke

Smoke outputs stay under DO NOT COMMIT:

```
python run_tsp_operator_suite.py --config configs/tsp_phase7_suite/smoke_suite.json --skip-judge
python aggregate_tsp_operator_runs.py --runs-root "DO NOT COMMIT/tsp_phase7_suite/smoke"
```

Official Campaign

The recommended official campaign uses 20 paired offsets across all conditions.

```
$replicates = @(
    @{ offset = 0; label = "a" },
    @{ offset = 1000; label = "b" },
    @{ offset = 2000; label = "c" },
    @{ offset = 3000; label = "d" },
    @{ offset = 4000; label = "e" },
    @{ offset = 5000; label = "f" },
    @{ offset = 6000; label = "g" },
    @{ offset = 7000; label = "h" },
    @{ offset = 8000; label = "i" },
    @{ offset = 9000; label = "j" },
    @{ offset = 10000; label = "k" },
    @{ offset = 11000; label = "l" },
    @{ offset = 12000; label = "m" },
    @{ offset = 13000; label = "n" },
    @{ offset = 14000; label = "o" },
    @{ offset = 15000; label = "p" },
    @{ offset = 16000; label = "q" },
    @{ offset = 17000; label = "r" },
    @{ offset = 18000; label = "s" },
    @{ offset = 19000; label = "t" }
)

python prepare_tsp_benchmarks.py
foreach ($r in $replicates) {
    python run_tsp_operator_suite.py --config configs/tsp_phase7_suite/01_operator_discovery.json
    --seed-offset $r.offset --replicate-label $r.label
}
python aggregate_tsp_operator_runs.py --runs-root runs/tsp_phase7_suite/operator_discovery
```

Outputs

Each run writes:

- suite_summary.json
- run_metadata.json
- report.md

- report.pdf

Each modular condition additionally writes:

- operator_report.json
- operator_report.md

The aggregate pass writes:

- aggregate_summary.json
- aggregate_report.md
- aggregate_report.pdf